Kate Scarboro

ENTRY-LEVEL SOFTWARE ENGINEER | AGENTIC AI, CLOUD & DATA SYSTEMS

Charlotte, NC | (919) 805-6711 | scarboro.kate@gmail.com | https://kate.is-a.dev | https://github.com/katenip

Professional Summary

M.S. Computer Science candidate (December 2026) with hands-on projects in agentic AI, local LLMs, cloud/Linux systems, applied ML, and GPU computing. Connects open-weight models to coding agents through MCP, using prompt/context engineering and human-reviewed testing. Seeking an entry-level software engineering role in AI products, cloud platforms, or developer tools.

Education

University of North Carolina at Charlotte
Master of Science in Computer Science | Expected December 2026 | GPA: **3.87**
Bachelor of Science in Computer Science | 2024

Technical Skills

Programming: Python, C#, SQL, JavaScript | **AI / Agents:** local LLMs, Ollama, Qwen, GPT-OSS, Cline, OpenCode, agent orchestration, MCP, tool calling, prompt/context engineering, model evaluation, human review
Cloud / Systems: AWS EC2, Oracle Cloud (OCI), Ubuntu, Linux CLI, SSH, server administration, virtualization | **Data / ML:** pandas, NumPy, scikit-learn, feature engineering, classification, A/B testing, Snowflake, Power BI, DAX | **Computing:** CUDA, parallel processing, GPU computing, Git/GitHub, Unity

Project Experience

Agentic AI Development Workflow

Independent project | 2026

- Configured local LLM inference with Ollama (Qwen and GPT-OSS), evaluating model selection, context windows, GPU use, and coding response quality.
- Connected Cline and OpenCode coding agents to Unity and Roblox Studio through MCP servers for project inspection and editor actions.
- Used prompt/context engineering, bounded planning, tool calling, human review, and iterative testing to assess generated code changes.

Cloud Compute and Linux Server Operations

Personal infrastructure and cloud coursework | 2026

- Provisioned AWS EC2 and Oracle Cloud Infrastructure instances; administer Ubuntu through SSH and Linux CLI, managing services, logs, processes, and troubleshooting.
- Maintain AMP-managed game servers, covering instance configuration, service operation, and routine maintenance.

CUDA N-Body Simulation

C++ / CUDA | Parallel and GPU Computing coursework

- Implemented and benchmarked a GPU-accelerated simulation on an NVIDIA RTX 3080, scaling experiments to 8,000 bodies across 800 simulation steps.
- Compared pageable and pinned host/device memory transfers and recorded bandwidth results to examine data-movement costs.

Food Inspection Risk Analytics

Power BI, DAX, Python | Visual Analytics and ML projects

- Built a Mecklenburg County restaurant inspection dashboard with DAX risk measures and maps to show current performance and predicted decline.
- Preparing inspection and violation histories for Python classification; engineering features to identify higher-risk establishments.

Unity Game Systems

C# / Unity | Selected projects

- Developed SuperPuzzleBar, a playable 2D puzzle-platformer prototype featuring time manipulation and ghost replay; built SpaceBugBlast, a roguelike shooter with procedural levels and dynamic spawns.

Training and Professional Development

AWS Academy Data Engineering (in progress); AWS certification preparation. DataCamp: A/B Testing in Python; Introduction to GitHub Concepts; Machine Learning with PySpark; Explainable AI in Python; Introduction to Regression with statsmodels in Python.

Experience

Online Math Tutor | Thinking Feet, Remote | Sep 2023 - Jul 2024
Adapted explanations to student needs and broke technical concepts into clear, step-by-step instruction.
Batch Cook | CMS Cafeteria, Charlotte, NC | Aug 2024 - Jun 2025
Coordinated timing across multiple tasks in a fast-paced service environment.